

2. BIOLOGICAL CLASSIFICATION

- I. Many attempts have been made to classify living organisms
 - II. Classification was done not using scientific criteria
 - III. **Aristotle** (350 BC) was the earliest to attempt a more scientific basis for classification.
 - He used simple morphological characters to classify plants into trees, shrubs and herbs
 - He also divided animals into two groups those with had red blood (Enaima) and those that did not (Anaima)
 - He also divided based on habitat into Aquatic (water) and terrestrial (land).
- In **Linnaeus** time (1753) a **two kingdom classification** with **Plantae** and **Animalia** developed that included all plants and animals

Disadvantage of two kingdom classification

- This system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular and photosynthetic (green algae) and non- photosynthetic (fungi)
- Large number of organisms did not fall into either category
- Hence the two kingdom classification used for a long time was found inadequate.

Extra Information on Demerits of Two Kingdom Classification

- Earlier classification system included bacteria, cyanobacteria (blue green algae), fungi, mosses, ferns, gymnosperms, and angiosperms under 'Plants' based on the presence of cell wall
- It brought together the prokaryotic bacteria and blue green algae with other group which were eukaryotic
- It also grouped together the unicellular organisms and multicellular ones, for example *Chlamydomonas* and *Spirogyra* were placed together under algae
- The classification did not differentiate between heterotrophic group fungi and the autotrophic green plants. Though they also showed a characteristic difference in their cell wall composition. The fungi had chitin in their wall, while the green plants had a cellulosic cell wall.
- This placed together the groups which widely differed in other characteristics.
- A need was also felt for including beside gross morphology, other characteristics like
 - Cell structure
 - Nature of cell wall
 - Mode of nutrition
 - Habitat
 - Methods of reproduction
 - Evolutionary relationship
- Classification systems for the living organisms hence undergone several changes over time.

Five kingdom classification

- It is the advanced and most popular system of classification
- Proposed by **R. H. Whittaker** (1969)

Criteria for five kingdom classification

- Cell structure,
- Thallus organization,
- Mode of nutrition reproduction
- Phylogenetic relationships.

Characteristics of the Five Kingdoms

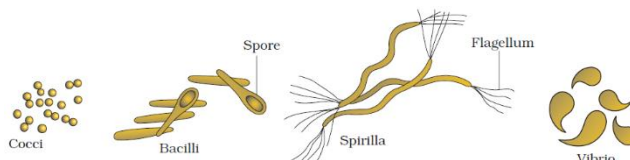
Characters	Five kingdoms				
	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellular (polysaccharide + amino acid)	Present in some	Present (without cellulose)	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organization	Cellular	Cellular	Multicellular / loose tissue	Tissue/ organ	Tissue/ organ / organ system
Mode of nutrition	Autotrophic (chemosynthetic and photosynthetic) and Heterotrophic (saprophytic / parasitic)	Autotrophic (photosynthetic) and heterotrophic	Heterotrophic (saprophytic / parasitic)	Autotrophic (photosynthetic)	Heterotrophic (Holozoic / Saprophytic etc)

Merits of Five kingdom classification

- All prokaryotic organisms were grouped together under kingdom Monera
- The unicellular eukaryotic organisms were placed in kingdom Protista which brought together *Chlamydomonas*, *Chlorella* (earlier placed in algae within plants and both having cell walls) with *Paramecium* and *Amoeba* (which were earlier placed in the animal kingdom which lack cell wall).
- It has put together organisms which in earlier classification were placed in different kingdom.

I. Kingdom Monera (All prokaryotes)

- Bacteria are sole members of this kingdom
- Most abundant microorganisms
- Occur almost everywhere including extreme environments such as **hot springs, deep oceans, snow, and deserts**
- Many live in or on other organisms as parasites
- Flagella when present comprise of protein flagellin
- Cell lacks membrane bound cell organelles.
- Cell wall when present made up of peptidoglycans
- Can be classified in four groups based on shape – the spherical Coccus (pl.: cocci), the rod-shaped Bacillus (pl.: bacilli), the comma-shaped Vibrium (pl.: vibrio) and the spiral Spirillum (pl.: spirilla)
- Cell structure is very simple but complex in behaviour
- Show the most extensive metabolic diversity
- Based on mode of nutrition, bacteria can be of following types
 - Autotrophic – Chemosynthetic (uses inorganic substrates) and Photosynthetic (Use CO₂ and H₂O)
 - Heterotrophic – Parasitic and Saprophytic.



1. Archaeobacteria (Living Fossils) (Primitive bacteria)

- Unique since they live in the most harsh habitats
- They are mainly Anaerobes (Live in absence of oxygen)
- Bacteria based on habitat can be named as
 - Halophiles (Salty areas)
 - Thermoacidophiles (Hot springs)
 - Methanogens (Marshy areas)
- Different from other bacteria in having a different cell wall structure which provides capacity to survive in these extreme conditions
- Methanogens present in the guts of ruminants are responsible for methane (biogas) production from their dung.

2. Eubacteria (True Bacteria)

- Characteristic feature – presence of rigid cell wall and if motile, a flagellum
- In Eubacteria there are two types –
 - I) Autotrophs/Autotrophic
 - i. Photosynthetic autotrophs – Cyanobacteria (blue green algae)
 - ii. Chemosynthetic autotrophs
 - II) Heterotrophs/Heterotrophic (decomposers)

I) Autotrophs / Autotrophic

i) Photosynthetic autotrophs

◆ Cyanobacteria (Blue green Algae)

- Habitat - marine or terrestrial.
- Have chlorophyll *a* similar to green plants; hence, they are photosynthetic autotrophs
- Unicellular, colonial or filamentous,
- Colonies surrounded by gelatinous sheath
- Some can fix atmospheric nitrogen in specialized cells called, **heterocysts**.

Example – *Nostoc* and *Anabaena*

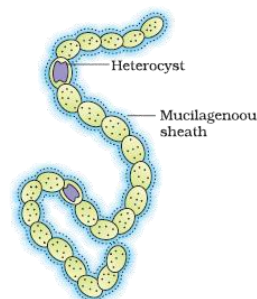


Figure: A filamentous blue-green algae – *Nostoc*

ii) Chemosynthetic Autotrophs

- Oxidize various inorganic substances such as nitrites, nitrates, ammonia, etc.
- Play key role in recycling nutrients such as nitrogen, sulphur, iron, Phosphorus etc.

II) Heterotrophic Bacteria (Decomposer)

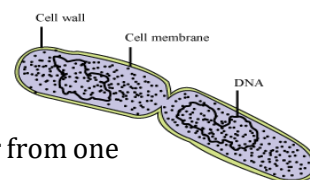
- Most abundant in nature
- Important decomposers
- Helpful in making curd from milk, production of antibiotics, fixing nitrogen in legumes
- Some are pathogens, causing damage to humans, crops, cattle, and pets.
- In humans, they cause diseases such as cholera (*Vibrio cholerae*), typhoid (*Salmonella typhi*) while they cause citrus canker (*Xanthomonas axonopodis*) in crop plants and anthrax (*Bacillus anthracis*) in cattle.

Mycoplasma

- Smallest known living creature
- Devoid of cell wall
- Can survive without oxygen
- Many are pathogens of humans, animals, and plants.

Reproduction in Bacteria

- Bacteria Reproduce asexually mainly by fission
- Under unfavourable conditions – reproduction through spores
- Sexual reproduction may also take place in a special way by DNA transfer from one bacterium to another and this process is called as **conjugation**.



II. Kingdom Protista

General Characteristics

- Contains all **unicellular eukaryotes**
- Primarily aquatic
- Cell body contains well-defined nucleus and other membrane-bound organelles
- Flagella and cilia when present, made up of protein tubulin.
- Reproduce asexually and sexually, by a process involving cell fusion and zygote formation

Classes of Protista

1. Chrysophytes (Planktons)
2. Dinoflagellates
3. Euglenoids
4. Slime moulds
5. Protozoans

1. Chrysophytes (Planktons)

- Microscopic; float passively in water currents
- Found in fresh water as well as marine environments
- Mostly photosynthetic
- Includes diatoms and golden algae (desmids)

Diatoms

- Cell wall forms two thin overlapping shells
- Walls embedded with silica, and thus, indestructible
- Fossilised remains of diatoms are referred to as 'diatomaceous earth'.
- Diatomaceous earth is used for polishing, filtration of oils and syrups and making sound and fire proof rooms
- Diatoms are the chief 'producers' of the ocean

2. Dinoflagellates

- Mostly marine and photosynthetic
- Appear yellow, green, brown, blue or red, depending on pigments present in their cells
- Cell wall has stiff cellulosic plates on the outer surface
- Most of these have two flagella—one lies longitudinally and the other transversely, in a furrow between the wall plates
- Red dinoflagellates (Example – *Gonyaulax*) rapidly multiply and make the sea appear red (red tides)
- Toxins released by these organisms may kill other marine animals

3. Euglenoids

- Majority are fresh-water organisms.
- Found in stagnant water.
- Have protein-rich layer called **pellicle**, instead of cell wall, which makes their body flexible
- They have 2 flagella, a short and a long one.
- In the presence of sunlight, they behave like autotrophs; in the absence of sunlight, they behave like heterotrophs, predating on smaller organisms. Hence mixotrophic nutrition.
- Pigments are similar to those present in higher plants.
- They are connecting link between plants and animals

4. Slime Moulds

- Saprophytic protists
- Move along decaying twigs and leaves
- Under suitable conditions, they form an aggregation called plasmodium, which grows and spreads
- Under unfavourable conditions, plasmodia differentiate to form fruiting bodies bearing spores at their tips
- Spores possess extremely resistant true walls, which make them capable for surviving for a long time under adverse conditions
- Spores disperse by air currents

5. Protozoans

- All are heterotrophs
- Live as predators or parasites
- Believed to be the primitive relatives of animals
- Types of Protozoans
 - a. Amoeboid protozoans
 - b. Flagellated protozoans
 - c. Ciliated protozoans
 - d. Sporozoans

a. Amoeboid Protozoans

- Live in freshwater, sea water or in moist soil
- Move and capture their prey by putting out pseudopodia (false foot); Example: *Amoeba*
- Marine forms have silica shells on their surface
- Some of them, such as *Entamoeba*, are internal parasites causes amoebic dysentery.

b. Flagellated Protozoans

- Either free-living or parasitic
- Bear flagella
- The parasitic forms cause diseases such as sleeping sickness; Example: *Trypanosoma*, *Giardia intestinalis*. Leishmania, causes leishmaniasis (black fever, Dum dum fever)

c. Ciliated Protozoans

- Aquatic and actively moving
- Show presence of thousands of cilia
- Have a cavity (gullet) which opens on the outside
- Movement of cilia causes food-laden water to enter the gullet
Example: *Paramecium*

d. Sporozoans

- Include diverse organisms with spore-like infectious stage in their life cycle
- Example: *Plasmodium*, which causes malaria

III. Kingdom Fungi

General Characters

- Constitute a unique kingdom of heterotrophic organisms
- Show great diversity in morphology and habitat
- Some are parasitic while most are saprophytic (feed on dead organisms).
- Except *Yeasts* (which are unicellular), other are multicellular and filamentous.
- The body consists of long, slender, thread-like structures called hyphae.
- A network of hyphae is known as mycelium.
- Some hyphae are continuous and multinucleate known as coenocytic hyphae while others have septatae (cross walls).
- Cell wall is composed of chitin and polysaccharides.
- Shows symbiotic association
 - with algae in **lichens**
 - with roots of higher plants in **mycorrhiza**
- Reproduction in fungi involves following methods
 - Vegetative means – by fragmentation, fission and budding
 - Asexual means – by spores called conidia or sporangiospores or zoospores.
 - Sexual means – by oospores, ascospores and basidiospores.
- Sexual reproduction involves
 - **plasmogamy** – fusion of protoplasm of two gametes
 - **karyogamy** – fusion of two nuclei
 - **meiosis in zygote** resulting in haploid spores
- During sexual reproduction in fungus,
 - fusion of two haploid hyphae of compatible mating types takes place which may result
 - in a diploid cell ($2n$) in some fungi or
 - an intervening dikaryotic stage ($n + n$) called dikaryon and a phase called dikaryophase occurs - mainly in classes Ascomycetes and Basidiomycetes

The dikaryotic phase of the fungal life cycle is unique. During this unusual phase, which is common in many species of fungi, cells contain two distinct nuclei. These two nuclei divide simultaneously as the mycelium grows; growth continues until fusion occurs during karyogamy resulting in diploid condition.

- Formation of fruiting bodies takes place
- Production of haploid spores occurs in fruiting bodies as a result of meiosis

Classes of Fungi

- Based on morphology of mycelium, mode of spore formation and fruiting bodies.
 1. Phycomycetes
 2. Ascomycetes
 3. Basidiomycetes
 4. Deuteromycetes

1. Phycomycetes (Algal fungi)

- Found in aquatic as well as damp habitats
 - Some are obligate parasites on plants
 - Mycelium – aseptate and coenocytic
 - Asexual reproduction – by motile zoospores or non-motile aplanospores
 - Spores produced endogenously in sporangium
 - Zygosporangia formed by fusion of two gametes which may be isogamous, anisogamous, or oogamous
- Example: *Mucor*, *Rhizopus* (Bread mould), *Albugo* (Plant parasite)

2. Ascomycetes (Sac fungi)

- Commonly called sac-fungi
- Unicellular (*Yeast*) or multicellular
- They are saprophytic, decomposers, parasitic or coprophilous (growing on dung)
- Mycelium – branched and septate
- Asexual spores (conidia) produced exogenously on special mycelium called conidiophores
- Sexual spores (Ascospores) produced exogenously in sac-like asci, arranged in different types of fruiting bodies called ascocarps
- Example: *Aspergillus*, *Claviceps*, *Neurospora* – Used extensively in biochemical and genetic work
- Several members such as morels, buffles and truffles are edible (considered as delicacies).

NOTE – Wonder drug – Penicillin is extracted from *Penicilium notatum* and *P. chrysogenum* (source of antibiotic)

3. Basidiomycetes (Club fungi)

- Commonly known members are mushrooms, toad stools, puff balls, or bracket fungi
- Grow in soil, on logs and tree stumps, or as plant-parasites (rusts and smuts)
- Mycelium – branched and septate
- Asexual spores absent and most common means of vegetative reproduction is fragmentation
- Sex organs absent, but plasmogamy takes place through fusion of vegetative cells of different strains
- Resultant structure is a dikaryon, which gives rise to basidium
- Karyogamy and meiosis occurs in basidium producing **four basidiospores**, which are produced exogenously on the basidia.
- Basidia are arranged in fruiting bodies called basidiocarps
- Example: *Agaricus* (mushroom), *Ustilago* (smut), *Puccinia* (rust)

4. Deuteromycetes (fungi imperfecti)

- Known as imperfect fungi because sexual phase is absent
- On recognition of sexual phase, the member is shifted to either class Ascomycetes or class Basidiomycetes
- Reproduce only by asexual spores (conidia)
- Mycelium is septate and branched.
- Some are saprophytic while others are parasitic.
- Being decomposers, these help in mineral cycling.
- Example: *Alternaria*, *Colletotrichum*, *Trichoderma*.

NOTE –

- *Alternaria solani* – causes early blight of potato
- *Phytophthora infestans* – causes late blight of potato
- *Colletotrichum* – causes bud or red rot of sugarcane

IV. Kingdom Plantae

General Characteristics

- Contains all eukaryotic-photosynthetic organisms
- Few are partially heterotrophic (insectivorous plants such as *Dionaea* (venus flytrap), Bladderwort, *Nepenthes* (pitcher plant), *Sarracenia* (a genus of 8 to 11 species of North American pitcher plant an insectivorous) *Drosera* (sun dew) or parasites (*Cuscuta*).
- Plant cells – eukaryotic with prominent chloroplasts and cellulosic cell wall
- Exhibit alternation of generation. i.e., plants life cycle consists of Diploid Sporophytic phase and haploid gametophytic phase
- Major divisions in kingdom plantae included are
 1. Algae
 2. Bryophyte
 3. Pteridophytes
 4. Gymnosperms
 5. Angiosperms

V. Kingdom Animalia

General Characteristics

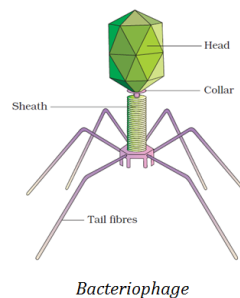
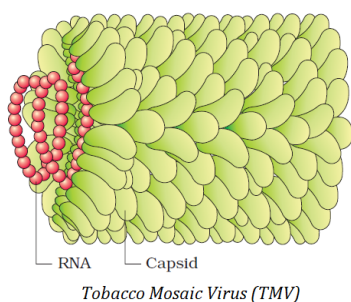
- Heterotrophic, eukaryotic, multicellular organisms
- Animal cells lack cell wall.
- Directly or indirectly depend on plants for food
- Digestion is internal and store food reserve as glycogen or fat.
- Follow definite growth pattern, grow into adults having definite shape and size
- Higher forms show elaborate sensory and neuromotor mechanism.
- Most capable of locomotion
- Sexual reproduction is by copulation of male and female followed by embryological development.

Viruses

- Being acellular, they are not placed anywhere in Whittaker's five kingdom classification.
- Viruses are not truly 'living' as they are living only inside the host body, else inert.
- They are non-cellular organisms characterized by inert crystalline structure outside the living cells.
- Obligate (compulsory) parasites
- Once inside a living cell, they take over the host cell machinery to replicate themselves, killing the host.
- Meaning of virus - venom or poisonous fluid
- Term coined by Pasteur
- D. J. Ivanowsky (1892) led the foundation of discovery of virus. He identified causative agent of Mosaic disease, which are smaller than bacteria as they can pass through bacteria-proof filters.
- M. W. Beijerinck (1898) found that the extract of infected plant causes disease in healthy plants. He called the fluid as *Contagium vivum fluidum* (infectious living fluid).
- W. M. Stanley (1935) crystallized viruses and demonstrated that crystals contain largely proteins, inert outside the host cell.

Structure of Viruses

- In addition to proteins, viruses also contain genetic material, either **DNA** or **RNA**, never both.
- It is a nucleoprotein and the genetic material is infectious.
- Based on host cell viruses can be
 1. Plant viruses – contains single stranded RNA – Ex- Tobacco Mosaic Virus (TMV)
 2. Bacterial viruses (bacteriophage) – contains double stranded DNA
 3. Animal viruses – contains double or single stranded RNA and double stranded DNA



- The protein coat is called **capsid**. It protects the nucleic acid and is made up of small subunits called **capsomeres**, which are arranged in helical or polyhedral geometric forms.
- Viruses cause diseases such as mumps, small pox, herpes influenza, and AIDS in humans.
- In plants, they cause mosaic, leaf rolling and curling, yellowing, vein clearing, dwarfing and stunted growth.

NOTE –

- **Virions** – completely assembled virus i.e., protein coat and nucleic acid outside the host.

Viroids

- Discovered by **T.O. Diener** in 1971
- Smaller than viruses and responsible for **spindle fibre disease** in potato (caused by PSTVD – Potatospindle tuber viroid and transmitted by Aphids) (all potatoes and tomatoes are susceptible to PSTVD) and **Chrysanthemum stunt disease**.
- Unlike viruses, they lack protein coat and exist as free RNA, hence named viroids.
- Viroid RNA is of low molecular weight (10^5 daltons) ($1 \text{ dalton} = 1.66 \times 10^{-27} \text{ kg}$)

Prions

- Prions are the infectious abnormally folded protein which caused neurological diseases and are similar in size to viruses.
- Diseases caused by prions are bovine spongiform encephalopathy (BSE) commonly called mad cow disease in cattle and its analogous variant Creutzfeldt–Jacob disease (CJD) in humans.

Lichens

- Symbiotic associations between fungi and algae
- Algal component – phycobiont (autotrophic) – Green algae or cyanobacteria
- Fungal component – mycobiont (heterotrophic) – Ascomycetes or Basidiomycetes
- Algae prepare food for fungi; in return, fungi absorb water and minerals and provide shelter to the algae
- Very good pollution indicators – do not grow in polluted areas